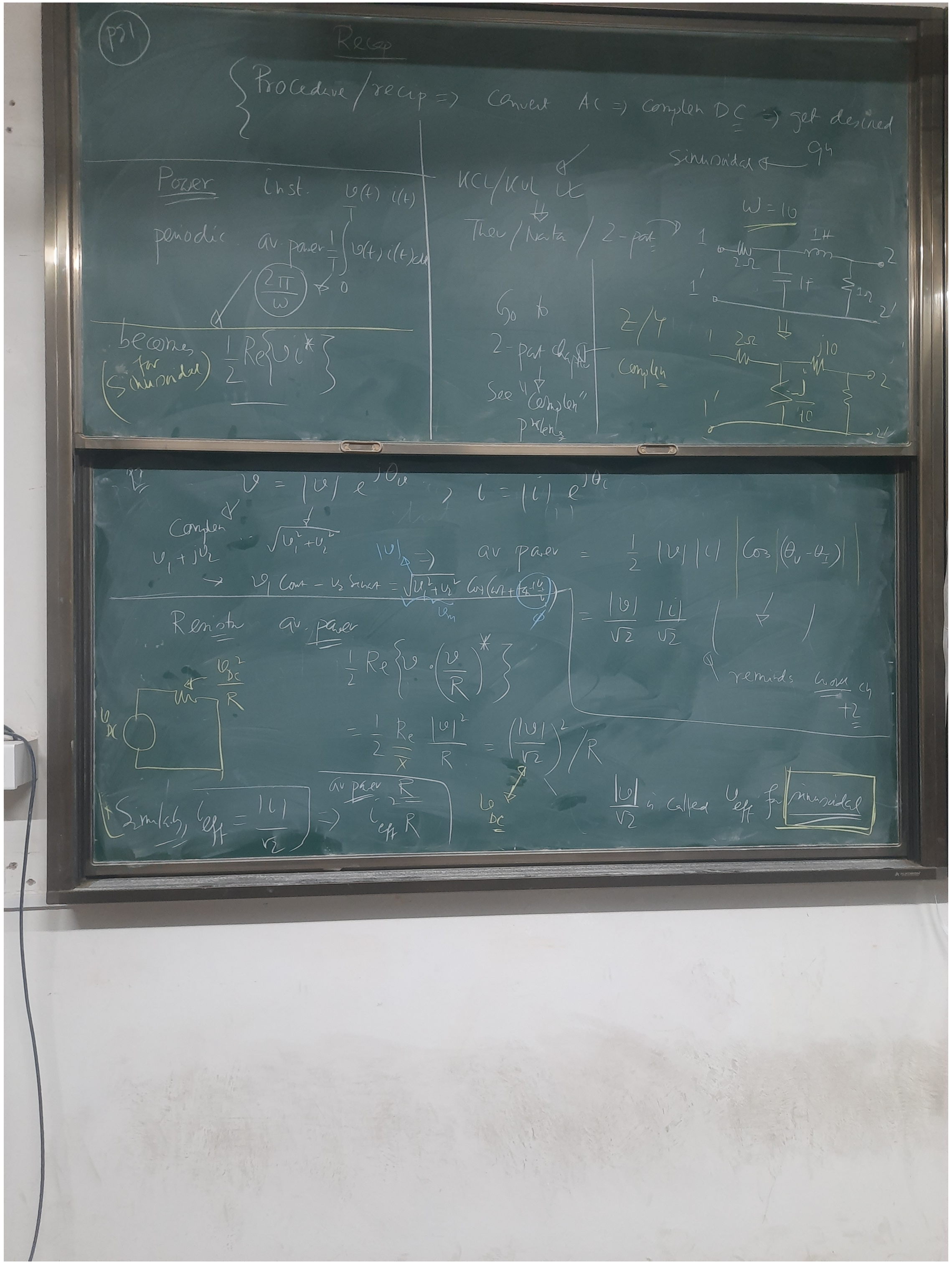


P31

Recap

{Procedure / recap ⇒ Convert AC ⇒ complex DC ⇒ get desired sinusoidal ← qn

Power

inst. $v(t)\,i(t)$

periodic av. power $\frac{1}{T}\int_0^T v(t)\,i(t)\,dt$, $T = \frac{2\pi}{\omega}$

becomes (for sinusoidal) $\frac{1}{2}\mathrm{Re}\{v\,i^*\}$

KCL/KVL etc

⇓

Thev / Norton / 2-port →

Go to 2-port chapter

↓

See "complex" problems

Z/Y

complex

$v = |v|\,e^{j\theta_v}$, $i = |i|\,e^{j\theta_i}$

complex $v_1 + jv_2$ → $|v| = \sqrt{v_1^2 + v_2^2}$

$\rightarrow v_1\cos\omega t - v_2\sin\omega t = \underbrace{\sqrt{v_1^2+v_2^2}}_{v_m}\cos(\omega t + \underbrace{\tan^{-1}\frac{v_2}{v_1}}_{\phi})$

av power $= \frac{1}{2}|v|\,|i|\,|\cos(\theta_v - \theta_i)|$

$= \frac{|v|}{\sqrt{2}}\,\frac{|i|}{\sqrt{2}}\,(\ \ )$

reminds [illegible] ch +2

Resistor av. power

$\frac{v_{DC}^2}{R}$

v_{DC}

$$\frac{1}{2}\mathrm{Re}\left\{v\cdot\left(\frac{v}{R}\right)^*\right\}$$

$$= \frac{1}{2}\cancel{\mathrm{Re}}\,\frac{|v|^2}{R} = \left(\frac{|v|}{\sqrt{2}}\right)^2 / R$$

v_{DC}

$\frac{|v|}{\sqrt{2}}$ is called v_{eff} for sinusoidal

Similarly, $i_{\text{eff}} = \frac{|i|}{\sqrt{2}}$ ⇒ av power $= i_{\text{eff}}^2 R$

v Sinusoidal

$$v_1 \cos\omega t + v_2(-\sin\omega t)$$

$$= \sqrt{v_1^2+v_2^2}\,\cos\left(\omega t + \tan^{-1}\frac{v_2}{v_1}\right)$$

In general,

$$v(t) = \boxed{v_m}\cos(\omega t + \phi)$$

Real

period $= \frac{2\pi}{\omega}$

Compare v_{rms} & v_{eff} (for sinusoidal)

$\Downarrow$

$$v_{rms} = v_{eff}$$

Why? $|v| = v_m$

(root) mean square

mean square

One period

$$\underbrace{\left(\frac{1}{T}\int_0^T v^2(t)\,dt\right)}_{\text{mean square}} \xrightarrow{\text{root}} \overset{v_{rms}}{\sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}} \overset{\text{sinusoidal}}{=} \frac{|v_m|}{\sqrt{2}}$$

PS3

PS4

Complex power

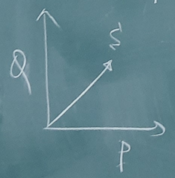
$$S = \frac{1}{2} v i^*$$

$$= \underbrace{\frac{1}{2}\mathrm{Re}(v i^*)}_{P} + j\,\underbrace{\frac{1}{2}\mathrm{Im}(v i^*)}_{Q}$$

Av power

reactive power

Power triangle



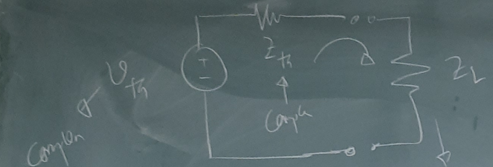
Flow from source ⇄ L/C

'reactive elements'

resistor cannot burn

pg5

Maximum power transfer



$Z_{th} = R_{th} + jX_{th}$

$Z_L = R_L + jX_L$

Av power $= \frac{1}{2}\mathrm{Re}\left(v_L\, i_L^*\right)$

$$= \frac{1}{2}\mathrm{Re}\left\{\underbrace{\frac{v_{th}\cdot Z_L}{Z_{th}+Z_L}}_{v_L}\underbrace{\left(\frac{v_{th}}{Z_L+Z_L}\right)^*}_{i_L}\right\} = \frac{1}{2}\mathrm{Re}\left\{\frac{|v_{th}|^2\cdot(R_L+jX_L)}{\boxed{|Z_L+Z_{th}|^2}}\right\}$$

$(R_L+R_{th})^2 + (X_{th}+X_L)^2$

$$(\text{args})\ \max_{Z_L}\ \frac{1}{2}\mathrm{Re}\left\{\frac{\cancel{|v_{th}|}\ \ R_L+jX_L}{(R_L+R_{th})^2+(X_L+X_{th})^2}\right\}$$

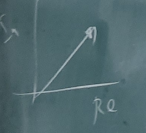
Z_L^*

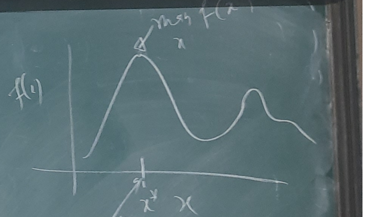
that max =

$$\max_{R_L,X_L}\ \mathrm{Re}\left\{\frac{R_L+jX_L}{(R_L+R_{th})^2+(X_L+X_{th})^2}\right\}$$

$$\max_{R_L,X_L}\ \boxed{\frac{R_L}{(R_L+R_{th})^2+(X_L+X_{th})^2}}$$

$$\max_{R_L}\left\{\max_{X_L}\left(\quad\right)\right\}$$ ← fn of R_L

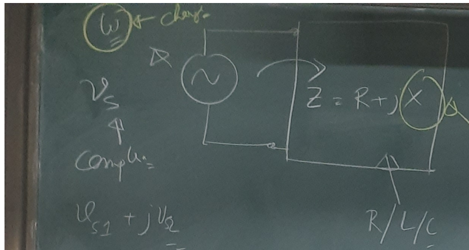




$$\arg\max_x\ 2x^2+3x$$

$$= 2\max_x\left(x^2+\frac{3x}{2}\right)$$

ω ← change



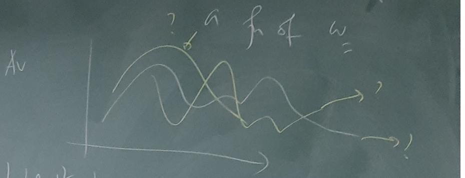
v_s ↓ complex

$v_{s1} + jv_{s2}$

change w/ "ω"

$j\omega L$ / $-1/\omega C$

Av power for source



$$\frac{1}{2}\mathrm{Re}\left\{v_s\left(\frac{v_s}{R+jX}\right)^*\right\} = |v_s|^2\,\frac{1}{2}\,\mathrm{Re}\left\{\frac{1}{R-jX}\right\}$$

maximizing ω??

$$= \frac{1}{2}\frac{|v_s|^2}{R^2+X^2}\cdot R \rightarrow \text{max when } X^2 = 0 \Downarrow X = 0$$

"ω" we are looking for ⟺ $X = 0$ ← fn of ω

PS2

PS8

$X = 0$ ⟺ v_s & i are in phase (phase diff = 0)

ω max power ⟺ ω for which $X = 0$ ⟺ v & i in phase
fn of ω

→ resonance & "ω" where it happens is called resonance freq.!